Business Problem:

DataCo Global sells clothing, sports and electronic products to customers in many countries. Management lacks a clear view of where its supply chain loses money and time. Many orders are delivered late, some orders are sold at a loss, and heavy discounts may reduce profit. Without clear analysis, the company cannot decide which shipping modes, regions and products need action.

This project analyzes DataCo's order, shipping and sales data to find the causes of late deliveries and low profit, and to support better operational decisions.

Objectives:

1. Measure overall performance: total sales, profit, order count and profit margin.
2. Find the late delivery rate and average delay by shipping mode, region and country.
3. Identify the top and bottom products and categories by sales and profit.
4. Find which orders lose money and what drives those losses, such as discount level, product or market.
5. Analyze sales and profit trends over time to find seasonal patterns.
6. Build a Power BI dashboard with four pages: Sales Overview, Delivery Performance, Profit and Loss Risk and Root Cause Analysis.
7. Give 5-6 data-based recommendations to reduce delays and losses.

Dataset Summary:

- **Source:** DataCo Smart Supply Chain for Big Data Analysis (Kaggle)
- **Period:** January 2015 to January 2018 (2018 is a partial year)
- **Tables:** 1 main table used (DataCoSupplyChainDataset.csv), loaded into PostgreSQL as dataco_orders, plus customers and products tables
- **Rows (raw):**
  - dataco_orders 180,519 (one row per order item)
  - Distinct orders about 66K
- **Columns:**
  - Raw: 53
  - Added: delay_days, is_late, is_loss, profit_margin, order_year, order_month, order_month_name
- **Key Features:**
  - Customer details (Customer ID, Segment, City, State, Country)
  - Order details (Order ID, Order Date, Order Status, Delivery Status, Market, Region, Country)
  - Shipping details (Shipping Mode, Real Shipping Days, Scheduled Shipping Days)
  - Product details (Product Name, Category, Department, Price)
  - Sales and profit (Sales, Quantity, Discount Rate, Profit per Order)
- **Missing Data:**
  - Customer zipcode, order zipcode: missing values, dropped(not needed)
  - Product description: empty, dropped
  - Customer Lname: missing values, dropped(not needed)

    - No other columns had missing values
- **Other files:**
    - DescriptionDataCoSupplyChain.csv: data dictionary
    - tokenized_access_logs.csv: website logs, not used (no key to link to orders)
- **Analysis scope:** 180,519 order items, about 173K delivered items (canceled shipments excluded for delivery analysis), 5 markets, 4 shipping modes.

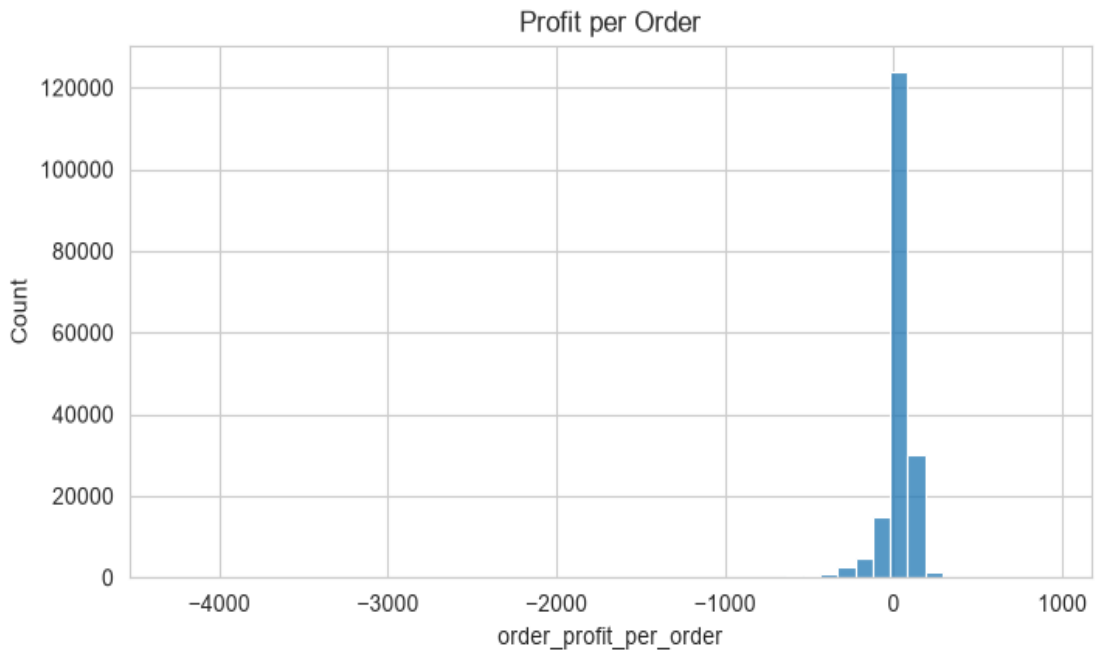
Exploratory Data Analysis using Python:

It starts with data preparation and cleaning in python.

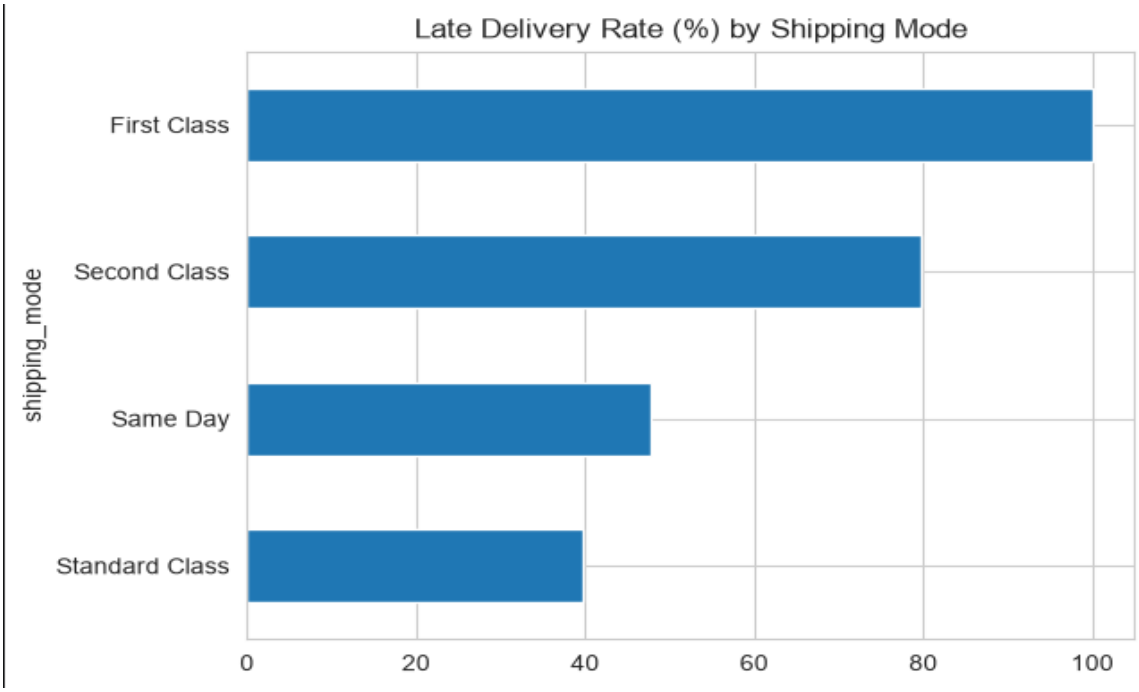
- **Data Loading:** Imported the dataset using pandas.
- **Exploration:** df.head() showed the first rows, df.info() showed the structure and data types, and df.describe() gave summary statistics. df.isnull().sum() counted the missing values in each column, and df.duplicated().sum() counted duplicate rows.
- **Dropping Columns:** Nine columns were removed from the dataset. customer_zipcode, order_zipcode and product_description had missing values, and product_description was empty in every row. customer_email, customer_password, customer_street, customer_fname and customer_lname are personal or private details, and product_image is only an image link. None of these columns was needed to answer the business questions, so removing them also reduced the size of the data and protected customer privacy.

Six plots were made in Python with matplotlib and seaborn to understand the data before building the SQL queries and the dashboard.

1. **Profit per order (histogram):** Shows how profit is spread across order items. Most orders earn a small profit, but a large share fall below zero, and a few extreme losses stand out as outliers.



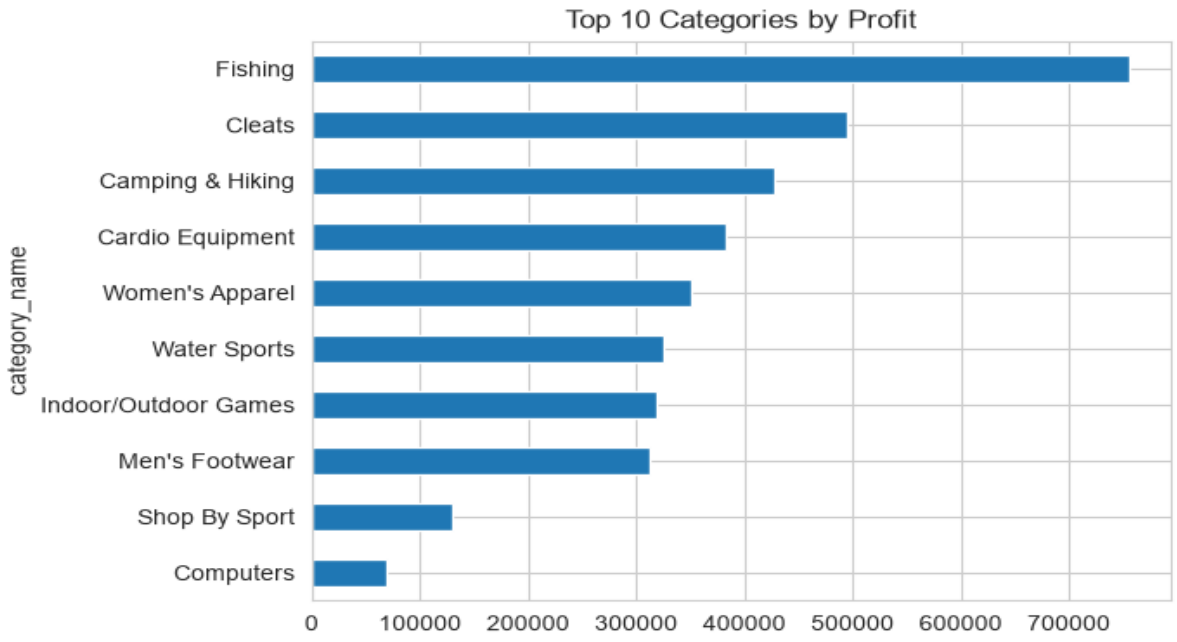
2. **Late delivery rate by shipping mode (bar chart):** Compares the share of late orders for each shipping mode. First Class had the highest late rate (100%), followed by Second Class (about 79%), while Same Day and Standard Class were lower.



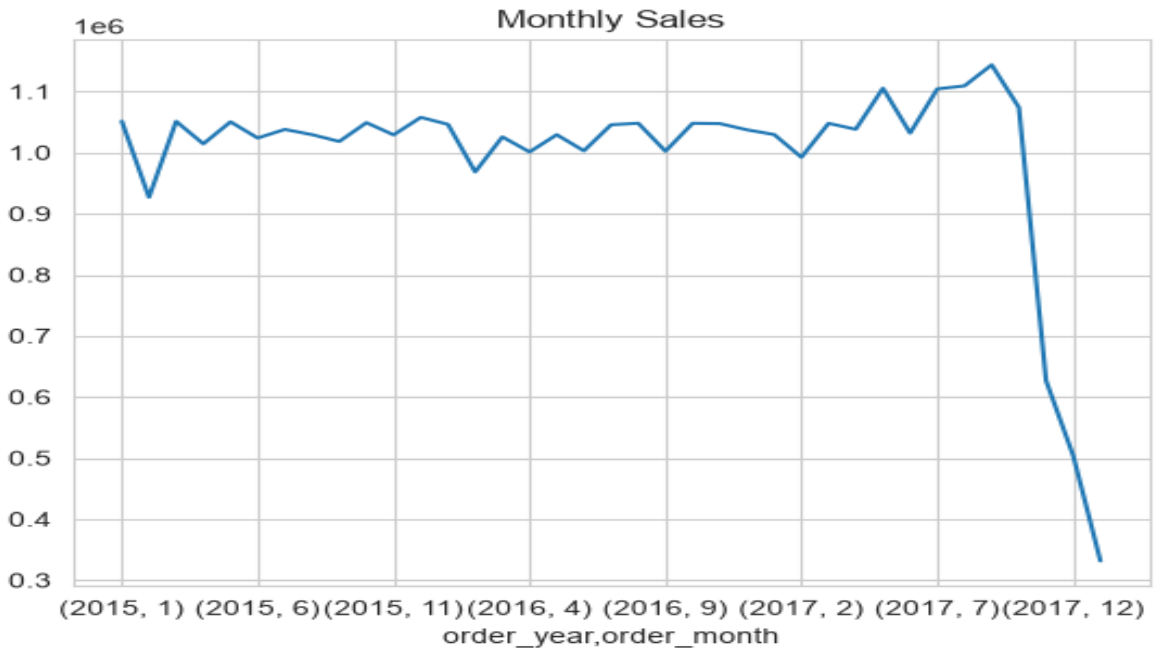
3. **Discount rate vs profit (scatter plot):** A sample of 5,000 orders was used to test whether higher discounts reduce profit. The plot showed no clear pattern. Losses appeared at every discount level, so discount alone does not explain the losses.



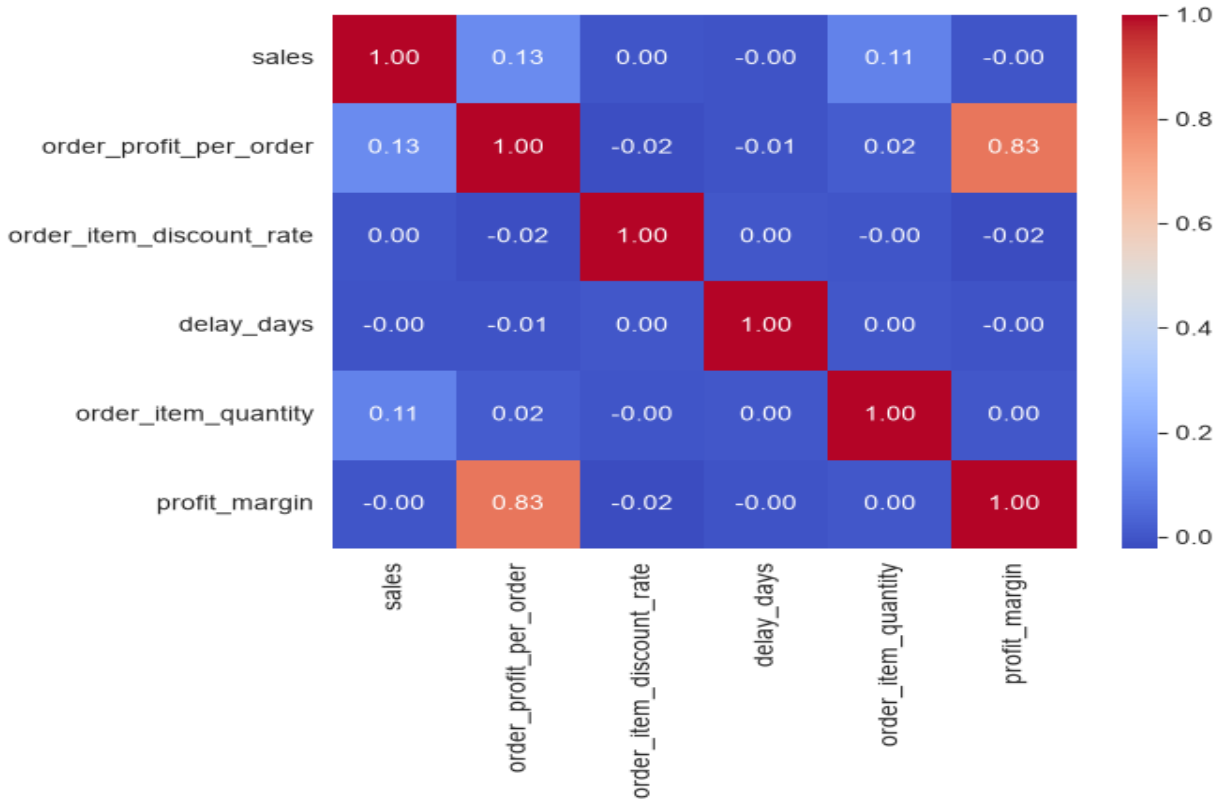
4. **Monthly sales trend (line chart):** Shows total sales for each month. Sales stayed fairly steady over the period, without a strong seasonal pattern.



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6. **Correlation heatmap:** The correlation heatmap showed that profit and profit margin are strongly correlated (0.83), which is expected because margin is calculated from profit. All other relationships were weak. Sales had a weak link with profit (0.13) and quantity (0.11). Discount rate (-0.02) and delay days (about 0) had almost no correlation with profit. This suggests that profit and loss are not driven by discount or delivery delay, and that categorical factors such as category, product and market need to be analyzed instead.



Database Integration: Connected Python script with postgresql and loaded the clean dataframe into the database for sql analysis.

Data Analysis Using SQL:

Database Design
The cleaned data was loaded into PostgreSQL as one flat table, dataco_orders, with one row per order item. Two lookup tables were then created from it to build a simple star-style model.

- **customers:** one row per customer, with the segment, city, state and country. DISTINCT ON (customer_id) keeps a single row for each customer ID, so customer details are not repeated on every order.
- **products:** one row per product, with the product name, category, department and price. DISTINCT ON (product_card_id) keeps a single row for each product.

Overall KPIs
This query calculates the main business totals. Total sales were $36.78M and total profit was $3.97M, giving a profit margin of 10.78%. The data holds about 180.5K order items in about 66K distinct orders. Order count uses COUNT(DISTINCT order_id), because one order can have several items. Profit margin is total profit divided by total sales, not an average of row margins.

total_sales	total_profit	order_items	order_count	profit_margin_pct
Filter...	Filter...	Filter...	Filter...	Filter...
36784735.01	3966902.97	180519	65752	10.78

Monthly Sales and Profit Trend

This query groups sales and profit by year and month, sorted in time order, to show how performance changes over time. Monthly sales stayed fairly steady at about $1M, without a strong seasonal pattern. The last year, 2018, contains only January, so it should not be compared with full years.

order_year	order_month	sales	profit
Filter...	Filter...	Filter...	Filter...
2015	1	1051590.08	111660.74
2015	2	927009.90	99140.66
2015	3	1051253.69	113778.21
2015	4	1014463.28	108083.68
2015	5	1050478.44	112147.90
2015	6	1024006.17	110147.16
2015	7	1038081.19	115624.06
2015	8	1029494.69	117979.77
2015	9	1018338.60	113467.94
2015	10	1049154.27	101757.87
2015	11	1029120.24	104427.36
2015	12	1057840.88	110641.55
2016	1	1046308.25	106780.95
2016	2	968542.85	86809.49
2016	3	1025853.12	101259.99
2016	4	1001211.58	114976.62
2016	5	1029400.20	110716.12
2016	6	1003059.55	104196.07
2016	7	1045715.62	119941.72
2016	8	1048200.25	110729.54
2016	9	1002397.04	123384.71
2016	10	1048130.55	116781.97
2016	11	1047590.14	109539.20
2016	12	1037408.17	105002.69
2017	1	1029698.02	114843.85
2017	2	992534.90	115111.07
2017	3	1048004.78	118688.90
2017	4	1038321.62	115961.06
2017	5	1105485.32	115014.64
2017	6	1032086.49	110399.29
2017	7	1104373.36	113026.70
2017	8	1109337.17	131501.16
2017	9	1143775.11	122462.39
2017	10	1073994.17	113447.17
2017	11	626914.38	67791.25
2017	12	503910.82	65837.63
2018	1	331650.12	33841.89

Top 10 Products by Profit

This query adds up the profit of each product and lists the 10 with the highest total. It shows which products contribute the most to company profit. Field & Stream Sportsman 16 Gun Fire Safe leads with $756220.77.Total profit favors products that sell in high volume, so a high-profit product can still have a low margin.

product_name	profit
Filter...	Filter...
Field & Stream Sportsman 16 Gun Fire Safe	756220.77
Perfect Fitness Perfect Rip Deck	493828.30
Diamondback Women's Serene Classic Comfort Bi	427455.57
Nike Men's Free 5.0+ Running Shoe	379915.82
Nike Men's Dri-FIT Victory Golf Polo	350421.03
Pelican Sunstream 100 Kayak	324076.37
O'Brien Men's Neoprene Life Vest	318451.43
Nike Men's CJ Elite 2 TD Football Cleat	311902.82
Under Armour Girls' Toddler Spine Surge Runni	126278.51
Dell Laptop	69656.81

Top 10 Categories by Profit

This query adds up profit for each product category and lists the 10 highest. It shows which product groups bring in the most profit. Fishing safe leads with $756220.77 and the second category with highest profit is cleats with $494636.92.The ranking shows where the business earns most of its profit, and it can guide stock and pricing decisions.

category_name	profit
Filter...	Filter...
Fishing	756220.77
Cleats	494636.92
Camping & Hiking	427455.57
Cardio Equipment	383011.10
Women's Apparel	350421.03
Water Sports	325146.96
Indoor/Outdoor Games	318451.43
Men's Footwear	311902.82
Shop By Sport	129813.96
Computers	69656.81

Bottom 10 Products by Profit

This query adds up the profit of each product and lists the 10 with the lowest totals. Only three products lost money overall: SOLE E35 Elliptical (-$965), Bushnell Pro X7 Jolt Slope Rangefinder (-$256) and SOLE E25 Elliptical (-$170), about -$1,391 in total. The other seven earned a small profit of $246 to $479. These losses are very small compared with the total profit of $3.97M, so the weakest products are not the main cause of lost profit.

product_name	profit
Filter...	Filter...
SOLE E35 Elliptical	-965.12
Bushnell Pro X7 Jolt Slope Rangefinder	-255.95
SOLE E25 Elliptical	-169.56
GoPro HERO3+ Black Edition Camera	245.63
Diamondback Girls' Clarity 24 Hybrid Bike 201	284.42
Polar Loop Activity Tracker	326.20
Cleveland Golf Women's 588 RTX CB Satin Chrom	370.61
CDs of rock	383.85
Garmin Forerunner 910XT GPS Watch	391.13
Stiga Master Series ST3100 Competition Indoor	479.28

Bottom 10 Categories by Profit

This query adds up profit for each product category and lists the 10 with the lowest totals. No category lost money overall. The lowest were Strength Training ($332), CDs ($384) and As Seen on TV! ($714), and the 10th lowest, Video Games, still earned $2,718. Together, these 10 categories earned about $13.1K, which is only about 0.3% of the total profit of $3.97M. So the weakest categories are small contributors, not a source of loss, and they may need a review of their pricing, costs or range only if they also have a low profit margin.

category_name	profit
Filter...	Filter...
Strength Training	332.31
CDs	383.85
As Seen on TV!	714.43
Books	883.01
Toys	900.71
Baby	1525.03
Golf Bags & Carts	1810.07
Basketball	1845.67
Men's Clothing	2006.04
Video Games	2717.52

Sales by Market

This query adds up sales for each market and ranks them from highest to lowest. Europe led with $10.87M (about 30% of total sales), followed closely by LATAM with $10.28M (28%). Together they made up about 57% of all sales. Pacific Asia contributed $8.27M (22%), USCA $5.07M (14%) and Africa $2.29M (6%). Sales are concentrated in Europe and LATAM, while Africa is the smallest market.

market	sales_market
Filter...	Filter...
Europe	10872396.80
LATAM	10277612.84
Pacific Asia	8273743.74
USCA	5066528.71
Africa	2294452.93

Sales by Region

This query adds up sales for each of the 23 order regions and ranks them from highest to lowest. Western Europe led with $5.89M (about 16% of total sales), followed by Central America with $5.67M (about 15%). Together they made up about 31% of all sales. South America ($2.96M) was third. The weakest regions were Central Asia ($0.11M), Canada ($0.19M) and Southern Africa ($0.23M), each under 1% of sales. Sales are concentrated in a few regions, and the top three account for about 40% of the total.

order_region	sales_region
Filter...	Filter...
Western Europe	5894380.77
Central America	5665712.10
South America	2960881.41
Northern Europe	2155830.65
Southern Europe	2047918.82
Oceania	2016654.20
Southeast Asia	1932495.57
Caribbean	1651019.33
West of USA	1571415.96
South Asia	1553680.92
Eastern Asia	1486401.34
East of USA	1371111.99
West Asia	1174671.78
US Center	1151355.77
South of USA	785783.95
Eastern Europe	774266.57
West Africa	727951.19
North Africa	634752.22
East Africa	376234.90
Central Africa	327263.02
Southern Africa	228251.59
Canada	186861.04
Central Asia	109839.93

Sales and Profit by Customer Segment

This query adds up sales and profit for each customer segment. Consumer is the largest segment, with $19.10M in sales (about 52% of the total) and $2.07M in profit. Corporate follows with $11.17M (about 30%) and $1.20M, and Home Office is the smallest with $6.52M (about 18%) and $0.69M. The

profit margins are very similar: about 10.9% for Consumer, 10.8% for Corporate and 10.6% for Home Office.

customer_segm...	sales_segment	profit_segment
Filter...	Filter...	Filter...
Consumer	19095790.16	2073487.67
Corporate	11168406.84	1202574.96
Home Office	6520538.02	690840.34

Late Delivery Rate by Shipping Mode

This query calculates the late delivery rate for each shipping mode, excluding canceled shipments. In total, 57.3% of delivered items arrived late. First Class had the highest rate, with all 26,513 items late (100%), followed by Second Class at 79.8%. Same Day was late 47.9% of the time and Standard Class 39.8%. Standard Class carries about 60% of the volume, so it still produces about 41% of all late items. First Class and Second Class together make up only 35% of the volume but account for about 54% of late deliveries.

shipping_mode	total_orders	late_deliveries	late_delivery_rate
Filter...	Filter...	Filter...	Filter...
First Class	26513	26513	100.00
Second Class	33806	26987	79.83
Same Day	9293	4454	47.93
Standard Class	103153	41023	39.77

Late Delivery Rate by Market

This query calculates the late delivery rate for each market, excluding canceled shipments. In total, 57.3% of delivered orders (98,977 of 172,765) arrived late. Europe had the highest rate at 57.69%, followed by USCA at 57.41% and Pacific Asia at 57.38%. LATAM was late 56.87% of the time and Africa 56.84%.

market	total_orders	late_deliveries	late_delivery_rate
Filter...	Filter...	Filter...	Filter...
Europe	48090	27743	57.69
USCA	24627	14138	57.41
Pacific Asia	39585	22712	57.38
LATAM	49309	28044	56.87
Africa	11154	6340	56.84

Late Delivery Rate by Order Region

This query calculates the late delivery rate for each order region, excluding canceled shipments. Late deliveries are widespread: 22 of the 23 regions have a late rate above 54%, against an overall rate of 57.3%. Central Africa had the highest rate at 60.15%, followed by Western Europe (58.53%) and South of USA (58.52%), while Canada had the lowest at 51.60%.

order_region	total_orders	late_deliveries	late_delivery_rate
Filter...	Filter...	Filter...	Filter...
Central Africa	1616	972	60.15
Western Europe	25867	15140	58.53
South of USA	3855	2256	58.52
South Asia	7455	4350	58.35
East of USA	6617	3849	58.17
East Africa	1781	1036	58.17
Southeast Asia	9136	5297	57.98
West Asia	5746	3322	57.81
Eastern Europe	3785	2182	57.65
US Center	5653	3252	57.53
South America	14184	8111	57.18
Central America	27174	15518	57.11
North Africa	3086	1762	57.10
Southern Europe	9030	5129	56.80
West of USA	7595	4313	56.79
Eastern Asia	6973	3955	56.72
Central Asia	542	306	56.46
Oceania	9733	5482	56.32
Northern Europe	9408	5292	56.25
Southern Africa	1100	617	56.09
Caribbean	7951	4415	55.53
West Africa	3571	1953	54.69
Canada	907	468	51.60

Average Delivery Delay by Shipping Mode

This query calculates the average delay in days for each shipping mode. Second Class had the longest average delay at 1.99 days, followed by First Class at 1.00 day and Same Day at 0.48 days, while Standard Class had none (0.00 days). The faster, more expensive modes perform worst against their promised delivery times.

shipping_mode	average_delay_d...
Filter...	Filter...
Second Class	1.99
First Class	1.00
Same Day	0.48
Standard Class	0.00

Top 10 Countries by Late Delivery Rate

This query calculates the late delivery rate for each country with more than 100 orders and lists the 10 highest. Ecuador had the highest rate at 69.39% (204 of 294 orders), followed by Níger (66.67%) and Mozambique (66.39%). All ten countries are well above the overall rate of 57.3%, ranging from 62.68% (Malasia) to 69.39%. Most of these countries have small order volumes (108 to 544 orders), so they have little effect on the company total.

order_country	total_orders	late_deliveries	late_delivery_rate
Filter...	Filter...	Filter...	Filter...
Ecuador	294	204	69.39
Níger	108	72	66.67
Mozambique	244	162	66.39
Irlanda	491	318	64.77
Trinidad y Tobago	138	89	64.49
Hungría	179	115	64.25
Somalia	141	90	63.83
Ghana	264	168	63.64
Angola	306	192	62.75
Malasia	544	341	62.68

Overall Loss-Making Orders

This query calculates how many orders resulted in a loss and what share of all orders they represent. Of 180,519 orders, 33,784 were loss-making, a loss order rate of 18.71%. This means almost one in five orders loses money, which makes loss-making orders a significant drag on overall profit.

total_orders	loss_order	loss_order_rate
Filter...	Filter...	Filter...
180519	33784	18.71

Top 10 Categories by Loss Order Rate

This query calculates the share of loss-making orders in each product category with more than 100 orders and lists the 10 highest. Men's Clothing had the highest loss order rate at 24.04% (50 of 208 orders), followed by Health and Beauty (23.48%) and Video Games (21.60%). All ten categories are above the overall loss order rate of 18.71%, ranging from 19.61% (Shop By Sport) to 24.04%. Shop By Sport is the largest category here, with 10,984 orders and 2,154 loss-making ones, so it has the biggest effect on total losses despite a lower rate.

category_name	total_orders	loss_order	loss_order_rate
Filter...	Filter...	Filter...	Filter...
Men's Clothing	208	50	24.04
Health and Beauty	362	85	23.48
Video Games	838	181	21.60
Golf Shoes	524	111	21.18
Pet Supplies	492	104	21.14
DVDs	483	102	21.12
Kids' Golf Clubs	384	77	20.05
Golf Apparel	441	88	19.95
Crafts	484	96	19.83
Shop By Sport	10984	2154	19.61

Loss Order Rate by Market

This query calculates the share of loss-making orders in each market. Pacific Asia had the highest loss order rate at 18.92% (7,806 of 41,260 orders), followed by Europe (18.75%) and USCA (18.65%), while Africa and LATAM were lowest at 18.58%. The gap between the best and worst market is only 0.34 percentage points, and all five markets sit close to the overall rate of 18.71%. This suggests loss-making orders are not a regional issue, and are more likely driven by product category, discounting or pricing.

market	total_orders	loss_order	loss_order_rate
Filter...	Filter...	Filter...	Filter...
Pacific Asia	41260	7806	18.92
Europe	50252	9420	18.75
USCA	25799	4812	18.65
Africa	11614	2158	18.58
LATAM	51594	9588	18.58

Top 10 Categories by Total Loss Amount

This query calculates the total money lost on loss-making orders in each product category and lists the 10 largest. Fishing lost the most at $728,570.95 across 3,209 loss-making orders, followed by Cleats ($452,594.21) and Camping & Hiking ($443,082.23). Together, these three categories account for about $1.62 million in losses. Fishing's loss is about 61% higher than that of Cleats, which ranks second. Computers had only 73 loss-making orders but still lost $74,967.13, about $1,027 per order, which suggests a few very large losses rather than frequent small ones.

category_name	loss_orders	total_loss_amount
Filter...	Filter...	Filter...
Fishing	3209	728570.95
Cleats	4590	452594.21
Camping & Hiking	2590	443082.23
Cardio Equipment	2332	402647.26
Water Sports	2924	334569.63
Women's Apparel	3923	323772.87
Men's Footwear	4169	309269.70
Indoor/Outdoor Games	3617	298637.48
Shop By Sport	2154	144310.56
Computers	73	74967.13

Loss Order Rate by Discount Rate

This query groups orders by discount rate and calculates the loss order rate and average profit ratio for each level. Loss rates stay almost flat across all 17 discount levels, ranging from 18.09% (9% discount) to 19.58% (20% discount), close to the overall rate of 18.71%. Average profit ratio is also stable at 0.11 to 0.13 at every level. The 0% discount group has a loss rate of 18.38%, almost the same as the 25% group at 18.48%. This suggests discounting is not the main driver of loss-making orders in this dataset, so the causes are more likely product cost, pricing or shipping.

discount_rate	total_orders	loss_orders	loss_rate_pct	avg_profit_ratio
Filter...	Filter...	Filter...	Filter...	Filter...
0.25	10029	1853	18.48	0.13
0.20	10029	1964	19.58	0.11
0.18	10029	1841	18.36	0.12
0.17	10029	1953	19.47	0.11
0.16	10029	1894	18.89	0.12
0.15	10029	1885	18.80	0.12
0.13	10029	1914	19.08	0.12
0.12	10029	1901	18.96	0.12
0.10	10029	1845	18.40	0.12
0.09	10029	1814	18.09	0.12
0.07	10029	1893	18.88	0.12
0.06	10029	1816	18.11	0.12
0.05	10029	1872	18.67	0.13
0.04	10029	1888	18.83	0.12
0.03	10029	1926	19.20	0.11
0.02	10028	1836	18.31	0.12
0.01	10028	1846	18.41	0.12
0.00	10028	1843	18.38	0.13

Top 3 Products by Profit in Each Category

This query ranks the products within each category by total profit and shows the top three. The most profitable products are Field & Stream Sportsman 16 Gun Fire Safe in Fishing ($756,220.77), Perfect Fitness Perfect Rip Deck in Cleats ($493,828.30) and Diamondback Women's Serene Classic Comfort bike in Camping & Hiking ($427,455.57). Profit is highly concentrated: in Water Sports, the Pelican Sunstream 100 Kayak earned $324,076.37 while the second product earned only $1,070.59, and in Indoor/Outdoor Games the top product earned $318,451.43. Many categories, such as Toys, Books, CDs and Baby, contain only one product. Two categories show loss-making products even within their top three: SOLE E35 Elliptical in Strength Training (-$965.12) and SOLE E25 Elliptical in Basketball (-$169.56). This means each category's profit depends heavily on a few key products.

category_name	product_name	profit	profit_rank
Filter...	Filter...	Filter...	Filter...
Accessories	Team Golf Texas Longhorns Putter Grip	3079.31	1
Accessories	Team Golf Pittsburgh Steelers Putter Grip	3039.53	2
Accessories	Team Golf St. Louis Cardinals Putter Grip	2815.12	3
As Seen on TV!	Nike Men's Free TR 5.0 TB Training Shoe	714.43	1
Baby	Baby sweater	1525.03	1
Baseball & Softball	adidas Men's F10 Messi TRX FG Soccer Cleat	8213.07	1
Baseball & Softball	adidas Kids' F5 Messi FG Soccer Cleat	2906.33	2
Baseball & Softball	adidas Brazuca 2014 Official Match Ball	1642.73	3
Basketball	Diamondback Boys' Insight 24 Performance Hybr	1730.81	1
Basketball	Diamondback Girls' Clarity 24 Hybrid Bike 201	284.42	2
Basketball	SOLE E25 Elliptical	-169.56	3
Books	Sports Books	883.01	1
Boxing & MMA	Under Armour Women's Micro G Skulpt Running S	4469.77	1
Boxing & MMA	Brooks Women's Ghost 6 Running Shoe	2606.85	2
Boxing & MMA	Nike Women's Free 5.0 TR FIT PRT 4 Training S	1564.75	3
Cameras	Web Camera	30289.80	1
Camping & Hiking	Diamondback Women's Serene Classic Comfort Bi	427455.57	1
Cardio Equipment	Nike Men's Free 5.0+ Running Shoe	379915.82	1
Cardio Equipment	Nike Women's Tempo Shorts	3095.28	2
CDs	CDs of rock	383.85	1
Children's Clothing	Children's heaters	27178.10	1
Cleats	Perfect Fitness Perfect Rip Deck	493828.30	1
Cleats	Total Gym 1400	808.62	2
Computers	Dell Laptop	69656.81	1
Consumer Electronics	Industrial consumer electronics	13223.40	1
Crafts	Porcelain crafts	25531.17	1
DVDs	DVDs	6655.43	1
Electronics	Titleist Pro V1x High Numbers Personalized Go	6671.09	1
Electronics	Titleist Pro V1 High Numbers Personalized Gol	6093.15	2
Electronics	Titleist Pro V1x High Numbers Golf Balls	4592.28	3
Fishing	Field & Stream Sportsman 16 Gun Fire Safe	756220.77	1
Fitness Accessories	Under Armour Hustle Storm Medium Duffle Bag	4067.61	1
Fitness Accessories	Bowflex SelectTech 1090 Dumbbells	1190.78	2
Garden	Lawn mower	33443.01	1
Girls' Apparel	adidas Youth Germany Black/Red Away Match Soc	8129.40	1
Girls' Apparel	TYR Boys' Team Digi Jammer	3658.98	2
Girls' Apparel	adidas Men's Germany Black Crest Away Tee	2833.72	3
Golf Apparel	Top Flite Women's 2014 XL Hybrid	1830.40	1

Month-over-Month Sales Trend

This query calculates total sales for each month and the change from the previous month, in dollars and

as a percentage. From January 2015 to October 2017, sales stayed stable at roughly $0.93M to $1.14M per month, with most monthly changes within ±7%. February was the weakest month in each year (-11.85% in 2015, -7.43% in 2016 and -3.61% in 2017) and was followed by a rebound in March. Sales peaked in September 2017 at $1,143,775.11. After that they fell sharply: -41.63% in November 2017 ($626,914.38), -19.62% in December and -34.18% in January 2018 ($331,650.12). This drop is unusually steep compared with the earlier pattern, so it may reflect incomplete data for the final months rather than a real decline in demand.

order_year	ordermonth	currentmonthsa...	previousmonths...	mom_change	mom_perc
Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
2015	1	1051590.08	NULL	NULL	NULL
2015	2	927009.90	1051590.08	-124580.18	-11.85
2015	3	1051253.69	927009.90	124243.79	13.40
2015	4	1014463.28	1051253.69	-36790.41	-3.50
2015	5	1050478.44	1014463.28	36015.16	3.55
2015	6	1024006.17	1050478.44	-26472.27	-2.52
2015	7	1038081.19	1024006.17	14075.02	1.37
2015	8	1029494.69	1038081.19	-8586.50	-0.83
2015	9	1018338.60	1029494.69	-11156.09	-1.08
2015	10	1049154.27	1018338.60	30815.67	3.03
2015	11	1029120.24	1049154.27	-20034.03	-1.91
2015	12	1057840.88	1029120.24	28720.64	2.79
2016	1	1046308.25	1057840.88	-11532.63	-1.09
2016	2	968542.85	1046308.25	-77765.40	-7.43
2016	3	1025853.12	968542.85	57310.27	5.92
2016	4	1001211.58	1025853.12	-24641.54	-2.40
2016	5	1029400.20	1001211.58	28188.62	2.82
2016	6	1003059.55	1029400.20	-26340.65	-2.56
2016	7	1045715.62	1003059.55	42656.07	4.25
2016	8	1048200.25	1045715.62	2484.63	0.24
2016	9	1002397.04	1048200.25	-45803.21	-4.37
2016	10	1048130.55	1002397.04	45733.51	4.56
2016	11	1047590.14	1048130.55	-540.41	-0.05
2016	12	1037408.17	1047590.14	-10181.97	-0.97
2017	1	1029698.02	1037408.17	-7710.15	-0.74
2017	2	992534.90	1029698.02	-37163.12	-3.61
2017	3	1048004.78	992534.90	55469.88	5.59
2017	4	1038321.62	1048004.78	-9683.16	-0.92
2017	5	1105485.32	1038321.62	67163.70	6.47
2017	6	1032086.49	1105485.32	-73398.83	-6.64
2017	7	1104373.36	1032086.49	72286.87	7.00
2017	8	1109337.17	1104373.36	4963.81	0.45
2017	9	1143775.11	1109337.17	34437.94	3.10
2017	10	1073994.17	1143775.11	-69780.94	-6.10
2017	11	626914.38	1073994.17	-447079.79	-41.63
2017	12	503910.82	626914.38	-123003.56	-19.62
2018	1	331650.12	503910.82	-172260.70	-34.18

Top 10 Customers by Total Sales

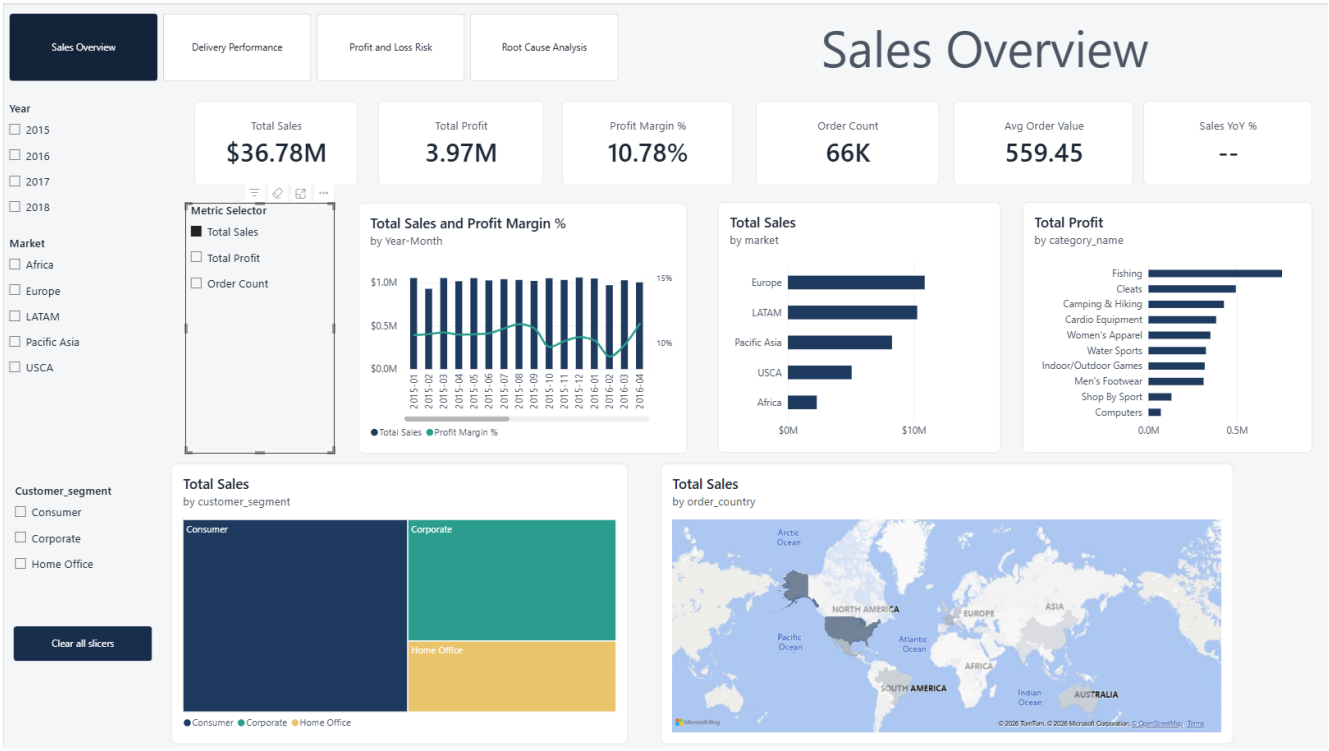
This query calculates total sales, order count and total profit for each customer and lists the 10 highest by sales. Customer 791 (Corporate, Canton, MI) had the highest sales at $10,524.17 across 13 orders,

but recorded a net loss of $866.38. Customer 9371 followed with $9,299.03, and customer 8766 with $9,296.14. Sales among the top ten are close, ranging from $8,761.98 to $10,524.17, and profit varies widely, from -$866.38 to $2,441.97 (customer 2641). Customer 1288 generated $9,019.11 in sales but only $403.11 in profit. Six of the ten customers are in Puerto Rico, five of them in Caguas, so a single city accounts for a large share of the top buyers.

customer_id	customer_segm...	customer_city	customer_state	customer_country	orders	total_sales	total_profit
Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
791	Corporate	Canton	MI	EE. UU.	13	10524.17	-866.38
9371	Consumer	Meridian	ID	EE. UU.	11	9299.03	1346.58
8766	Corporate	Caguas	PR	Puerto Rico	11	9296.14	1495.16
1657	Consumer	Caguas	PR	Puerto Rico	11	9223.71	2196.92
2641	Consumer	Carrollton	TX	EE. UU.	11	9130.92	2441.97
1288	Home Office	Caguas	PR	Puerto Rico	12	9019.11	403.11
3710	Consumer	Springfield	MO	EE. UU.	14	9019.10	1055.13
4249	Consumer	Caguas	PR	Puerto Rico	13	8918.85	439.71
5654	Home Office	Caguas	PR	Puerto Rico	15	8904.95	1045.36
5624	Consumer	Caguas	PR	Puerto Rico	12	8761.98	1178.15

Dashboards in Power BI:

Finally, built some dashboards to present insights visually.



Sales Overview | **Delivery Performance** | Profit and Loss Risk | Root Cause Analysis

Delivery Performance

Year
- 2015
- 2016
- 2017
- 2018

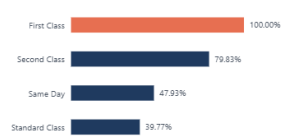
Market
- Africa
- Europe
- LATAM
- Pacific Asia
- USCA

Shipping_mode
- First Class
- Same Day
- Second Class
- Standard Class

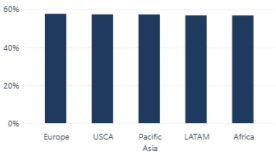
Clear all slicers

Late Delivery %	Avg Delay Days	Delivered Items	Late Items
57.29%	0.57	173K	99K

Late Delivery % by shipping_mode



Late Delivery % by market

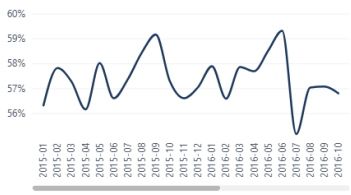


Late Delivery % by order_country



shipping_mode	Africa	Europe	LATAM	Pacific Asia	USCA	Total
First Class	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Same Day	46.37%	48.82%	50.36%	45.39%	46.43%	**47.93%**
Second Class	80.11%	80.21%	79.07%	80.38%	79.56%	**79.83%**
Standard Class	39.92%	39.76%	39.55%	40.03%	39.76%	**39.77%**
Total	**56.84%**	**57.69%**	**56.87%**	**57.38%**	**57.41%**	**57.29%**

Late Delivery % by Year-Month



order_country	Late Delivery %
Afganistán	60.00%
Alemania	58.80%
Angola	61.62%
Arabia Saudí	59.76%
Argelia	56.03%
Argentina	54.44%
Australia	56.70%
Austria	60.51%
Bangladés	59.96%
Barbados	57.02%
Bélgica	59.84%
Total	**57.28%**

Sales Overview | Delivery Performance | **Profit and Loss Risk** | Root Cause Analysis

Profit and Loss Risk

Clear all slicers

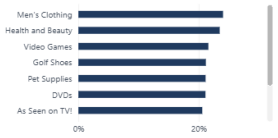
Year
- 2015
- 2016
- 2017
- 2018

Market
- Africa
- Europe
- LATAM
- Pacific Asia
- USCA

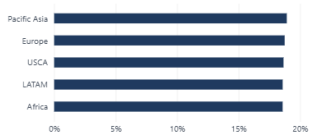
Category_name
- Accessories
- As Seen on TV!
- Baby
- Baseball & Softball
- Basketball
- Books
- Boxing & MMA
- Cameras
- Camping & Hiking
- Cardio Equipment

Total Loss Amount	Loss Order %	Loss Orders	Profit Margin %
$4M	18.7%	34K	10.78%

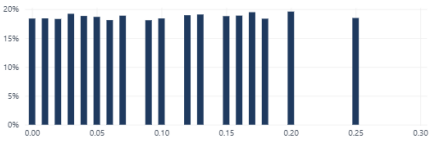
Loss Order % by category_name



Loss Order % by market



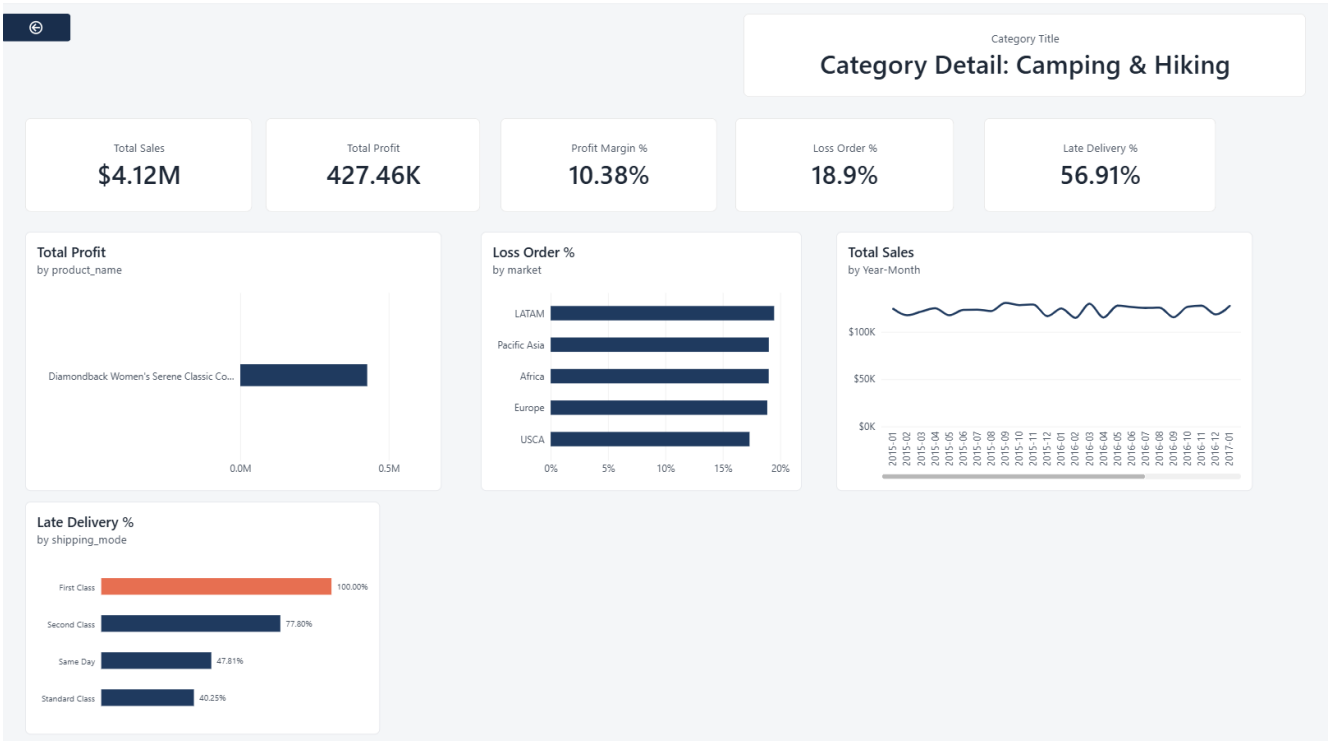
Loss Order % by order_item_discount_rate

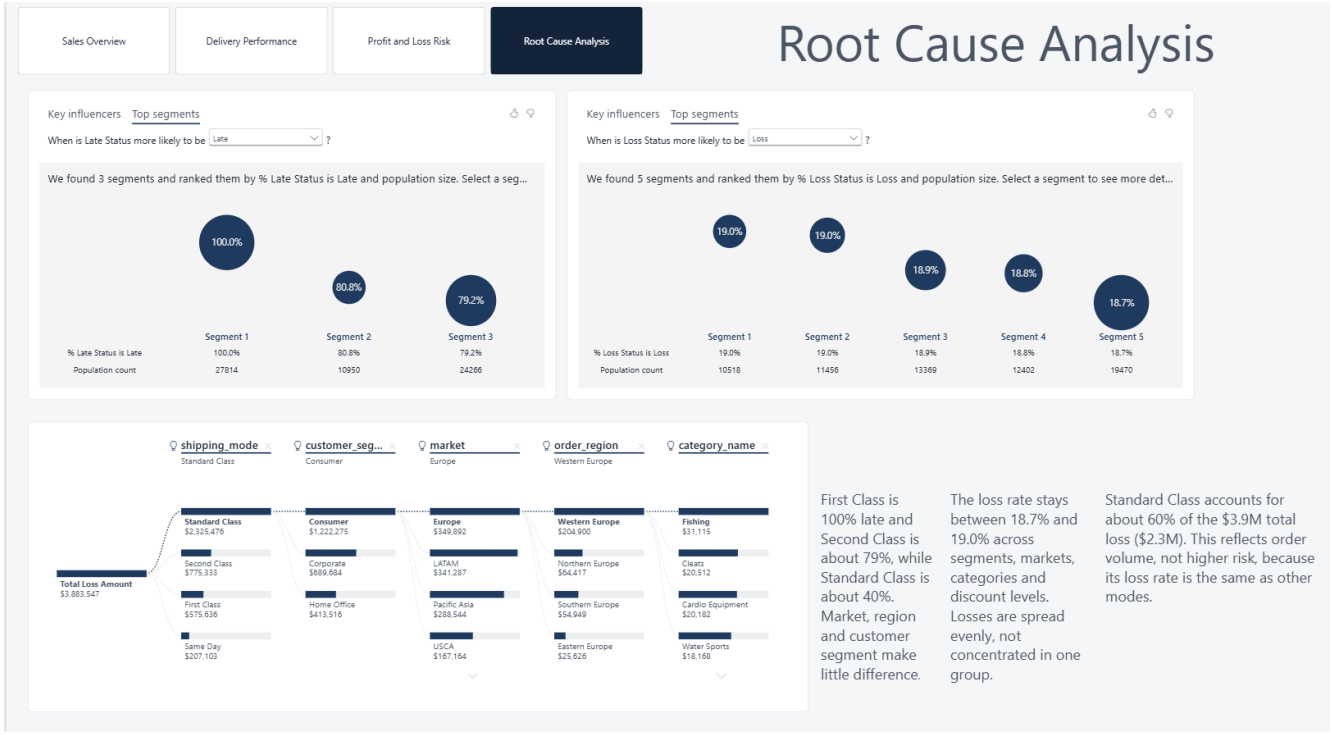


category_name	Africa	Europe	LATAM	Pacific Asia	USCA	Total
Accessories	16.9%	15.6%	16.8%	20.3%	19.4%	**17.9%**
As Seen on TV!		14.3%	30.8%			**20.6%**
Baby		9.8%		20.8%		**16.4%**
Baseball & Softball	20.0%	19.9%	15.3%	16.8%	13.7%	**16.8%**
Basketball		16.4%				**16.4%**
Books		17.9%		20.7%		**19.3%**
Boxing & MMA	25.8%	25.2%	16.5%	12.5%	13.0%	**18.9%**
Cameras		18.3%		17.6%		**18.2%**
Camping & Hiking	19.0%	18.9%	19.5%	19.0%	17.3%	**18.9%**
Cardio Equipment	16.5%	18.5%	18.2%	19.4%	19.9%	**18.7%**
CDs		12.0%		14.5%		**13.3%**
Children's Clothing		19.9%		16.2%		**19.2%**
Total	**18.6%**	**18.7%**	**18.6%**	**18.9%**	**18.7%**	**18.7%**

Top 10 Products by Loss Amount

product_name	category_name	Loss Orders	Loss Order %	Total Loss Amount
Field & Stream Sportsman 16 Gun Fire Safe	Fishing	3209	18.5%	$728,571
Perfect Fitness Perfect Rip Deck	Cleats	4583	18.7%	$450,978
Diamondback Women's Serene Classic Comfort Bi	Camping & Hiking	2590	18.9%	$443,082
Nike Men's Free 5.0+ Running Shoe	Cardio Equipment	2271	18.7%	$400,015
Pelican Sunstream 100 Kayak	Water Sports	2916	18.8%	$332,804
Nike Men's Dri-FIT Victory Golf Polo	Women's Apparel	3923	18.6%	$323,773
Nike Men's CJ Elite 2 TD Football Cleat	Men's Footwear	4169	18.7%	$309,270
O'Brien Men's Neoprene Life Vest	Indoor/Outdoor Games	3617	18.7%	$298,637
Under Armour Girls' Toddler Spine Surge Runni	Shop By Sport	2085	19.6%	$139,661
Dell Laptop	Computers	73	16.5%	$74,967





Business Recommendations

1. Fix First Class and Second Class delivery

- **Finding:** Premium shipping modes are the least reliable.

- **Evidence:** First Class was late on all 26,513 items (100%) and Second Class on 79.8%, with an average delay of 1.99 days. Together they carry 35% of the volume but 54% of all late deliveries.
- **Action:** Review the promised delivery times and the carrier performance for these two modes. Check first whether the 100% First Class result comes from its scheduled time being shorter than its actual time in this dataset.

2. Treat late delivery as a network-wide problem

- **Finding:** Late delivery is not limited to one market.
- **Evidence:** Late rates range from 56.84% (Africa) to 57.69% (Europe), and 22 of 23 regions are above 54%. Europe and LATAM make up about 57% of sales.
- **Action:** Improve delivery processes across the network, starting with Europe and LATAM, where the volume is highest.

3. Review pricing and costs of the top loss products

- **Finding:** Losses are concentrated by value in a few categories.
- **Evidence:** Fishing ($728,571), Cleats ($452,594) and Camping & Hiking ($443,082) lost $1.62M together. Loss rates are close to the 18.7% average in every market and category, so the cause is not one weak market. Computers lost about $1,027 per loss-making order, which points to a few very large losses.
- **Action:** Review the pricing, product cost and shipping cost of these categories, and investigate the largest single losses.

4. Do not rely on discount cuts to reduce losses

- **Finding:** Discounting does not explain loss-making orders.
- **Evidence:** The loss rate is 18.38% at 0% discount and 18.48% at 25%, and it stays between 18.09% and 19.58% at every level.
- **Action:** Focus on product cost, pricing and shipping instead of changing discount rules.

5. Test a small campaign for February

- **Finding:** February is the weakest month of the year.
- **Evidence:** Sales fell 11.85% in February 2015, 7.43% in 2016 and 3.61% in 2017, then rebounded in March.
- **Action:** Run a small pilot campaign in February and measure the change in sales. February has fewer days, so check sales per day before expanding.

6. Verify the sales drop after September 2017

- **Finding:** Sales fell sharply at the end of the data.
- **Evidence:** Sales fell 41.63% in November 2017 ($626,914), 19.62% in December and 34.18% in January 2018 ($331,650), after a steady period of about $0.93M to $1.14M per month.
- **Action:** Check the order counts and the last order date for these months. The drop may come from incomplete data, so confirm this before treating it as lost demand.